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Future of Immune Modulation in Animal Agriculture

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Keywords

veterinary immunology, immunomodulation, disease resistance, innate memory, trained innate immunity, tolerance, cytokine

Abstract

Immune modulation in animal agriculture has been of research interest for several decades, yet only a few immunomodulators have received regulatory approval in the United States and around the world. In this review, we summarize market and regulatory environments impacting commercial development of immunomodulators for use in livestock and poultry. In the United States, very few immunomodulators have received regulatory approval for use in livestock by either the US Department of Agriculture Center for Veterinary Biologics or the Food and Drug Administration (FDA). To date, only one immunomodulator has received FDA approval, and an extensive body of peer-reviewed literature is available regarding the basis for its use and health benefits. We present a more thorough review of the history and impact of this immune restorative. Finally, we discuss the interaction of immunomodulators on health, metabolism, and other factors impacting the future of immune modulation in livestock.

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1. INTRODUCTION

1.1. History of Immune Modulation in Animal Agriculture

A goal of immune modulation in animals raised for food production is to enhance resistance to infectious diseases. To achieve this goal, it is important to consider a few key concepts. An immunomodulator should do no harm to the animal. Immunomodulators can act to enhance an immune response or to suppress an immune response. Ideally, the homeostatic mechanisms of the animal maintain a balance in an immune response to avoid exceeding what is necessary to control an infection. However, there are many circumstances in the life of an animal in which immune capacity is suppressed or dysfunctional. Therefore, immunomodulators have been investigated for their ability to restore an impaired or dysfunctional immune status. In this situation, immunomodulation must not cause immunopathological harm to the host. Immunomodulators used in animals experiencing immune suppression will ideally restore their immune status back to an optimal level. Hence, the newborn period, weaning, comingling and transport stress, and the periparturient period are recognized time periods of increased incidence of intestinal and respiratory disease, as well as mastitis and uterine disease. These scenarios are frequent targets of research for restoring immune capacity impacted by stress. Another key aspect of immune modulation is that immunomodulators are not necessarily therapeutic products but instead best used as prophylactic disease management tools that require time to restore stress-impaired immune function. Thus, immunomodulators in this review are not vaccines, nor are they intended for use as antibiotic therapies; their disease prevention efficacy is exerted through immune-specific mechanisms effecting microbiological cures through enhanced immune mechanisms.

1.2. Immunomodulators for Animal Agriculture Fall under the Purview of Two Federal Regulatory Agencies

In the United States, two federal agencies have regulatory jurisdiction over approval and use of immunomodulators in animals. The original Pure Food and Drug Act of 1906 was passed by Congress and signed by President Theodore Roosevelt to prevent "manufacture, sale, or transportation of adulterated or misbranded or poisonous or deleterious food, drugs, medications, and liquors." Subsequently, the Viruses, Serums, Toxins, Antitoxins, and Analogous Products Act, known commonly as the Virus-Serum-Toxin Act, or VSTA (21 U.S.C. 151–159), was enacted in 1913. The latter was enacted primarily in response to substantial losses for hog producers from the unregulated manufacture and distribution of anti-hog cholera serum that was contaminated with virulent pathogens. The authority to license and regulate the production and trade of affected products was granted to the Secretary of Agriculture. The act states,

> It shall be unlawful for any person, firm or corporation to prepare, sell, barter, or exchange [in the United States], or ship or deliver for shipment in or from the United States. . .any worthless, contaminated, dangerous, or harmful virus, serum, toxin, or analogous product intended for use in the treatment of domestic animals.

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The delineation of the specific regulatory jurisdictions is explained more fully in **Supplemental Appendix 1.**

Internationally, regulatory approval for immunomodulators falls under different jurisdictions in various countries where they are licensed or approved for use. The European Medicines Agency (EMA) evaluates human and veterinary medicines in the European Union, Iceland, Norway, and Liechtenstein. One important caveat is that after Brexit—the United Kingdom's exit from the European Union—medications in the United Kingdom are no longer regulated by the EMA. The EMA is responsible for evaluating clinical data and providing drug recommendations based

on efficacy and safety information. Unlike US agencies, the EMA cannot authorize or approve certain medications. Instead, the EMA offers suggestions, and the European Commission provides the final regulatory decision. In Canada, regulatory authority falls with Health Canada, which is responsible for protecting human and animal health and the safety of Canada's food supply. Other countries have similar regulatory agencies with authority to approve the use of such compounds in animal agriculture. Although the regulatory environment is highly complex and can differ between and within countries, the intent of regulation is for compounds used in animal agriculture to be pure, safe, potent, and effective for their intended use. Finally, a company seeking regulatory approval in the United States and European Union can use the same studies to support both regulatory approvals if it can convince the agencies to agree on the clinical protocol designs and is willing to run studies in both geographical regions.

2. APPROVED IMMUNOMODULATORS FOR ANIMAL AGRICULTURE

Few immunomodulator products have received approval by regulatory agencies. They have distinct label claims and modulate immunity through distinct innate immune mechanisms.

2.1. Zylexis™

Zylexis™ was approved in 2005 in the United States for horses after almost a decade of use and patient data in Germany. The immunomodulator Zylexis™ contains inactivated *Parapoxvirus ovis* strain D1701 as the active substance and was initially licensed as Baypamun® (Bayer, Inc.). One of the earliest studies evaluated its immunomodulatory effects on reproductive performance of gilts following treatment and shipping transport stress (1). Authors reported increases in the number of gilts coming into estrus and conceiving after Baypamun® treatment. In addition, the cost of supportive medication was lower in treated versus untreated gilts. In vitro testing with pig leukocytes found no effect on phagocytosis, oxidative burst, or natural killer cell activity in swine; however, this study found enhanced release of interleukin-2, interferon-α (IFN-α), and IFN-γ, as well as strong lymphocyte proliferation (2). The effects on lymphoproliferative responses and IFN-γ production have differed in other studies with foals (3). A separate study in dairy cows found Baypamun® significantly reduced the prevalence of *Staphylococcus aureus* intramammary infections when administered prior to and just after calving (4). The authors suggested the effect was due to enhanced nonspecific resistance to infection. A later study found the inactivated *Parapoxvirus ovis* could possibly be used instead of antibiotics to reduce subclinical bovine mastitis, but the duration of effects was unknown (5).

This same product was acquired by another company and became known as Zylexis® (Pfizer AH, Inc.) in Europe. It was used as an aid in the prevention and treatment of infectious and/or stress-induced diseases by stimulating lymphocyte proliferation in dogs, cats, horses, cattle, and pigs (6). According to the manufacturer, Zylexis® is believed to work by inducing antiviral interferons and interleukins (e.g., IL-6, IL-12, TNF-α) in lymphocytes of various animal species. Innate immune system stimulation starts relatively quickly, within several hours to up to 2 days. The duration of action is approximately 10 to 14 days after the last injection. One study reported a much longer-term (4-year) effect on phagocytosis measurements in dogs (7). The manufacturer claims Zylexis® is useful against multifactorial infectious diseases and/or when infection pressure has increased. This broad application is achievable only due to its unique pathogen-nonspecific modulation of innate immunity (8).

Pfizer introduced Zylexis™ in the United States in 2005 for use in horses after almost a decade of use and patient data in Germany. Zylexis™ is reported to prime the equine immune system to function more efficiently against equine herpesvirus (EHV) types 1 and 4. EHV is

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highly contagious, and a large portion of horses in the United States are latent carriers of the virus. Latent infections may recrudesce when the host is stressed and lead to clinical disease. Zylexis™ is offered specifically as an aid in reducing upper-respiratory disease caused by these viruses and is considered safe in one-day-old foals up to adults, as well as in pregnant and lactating mares. Zylexis™-treated horses exhibited a reduction of clinical symptoms in a model of respiratory EHV infection. In vivo and in vitro data demonstrated that the use of Zylexis™ was safe and efficacious for the prophylaxis and treatment of stress-induced infectious diseases in horses. As an immunomodulator, Zylexis™ is not claimed to prevent a horse from contracting the infection or disease, but it can diminish the severity and duration of clinical symptoms (9). Globally, Zylexis™ is a unique immunomodulator indicated for the prevention and treatment of infectious and/or stress-induced diseases by stimulating the immune response in dogs, cats, horses, cattle, and pigs. Reviews of research on Zylexis™ in equine medicine are available (10–12). Curiously, a recent small study reported that Zylexis™ could induce significant innate antiviral immune responses that the authors suggested could be beneficial in controlling elephant endotheliotropic herpesviruses infections in Asian elephants (*Elephas maximus*) (13).

2.2. Zelnate™ Approved in 2015

According to the manufacturer (Bayer), Zelnate™ was approved as a US Department of Agriculture–Center for Veterinary Biologics (USDA-CVB)-approved immunomodulator based on technology developed by Juvaris BioTherapeutics. As such, it represented a new class of compound for bovine respiratory disease (BRD) as an immunomodulator; it is neither an antibiotic nor a vaccine. Antibiotics are used for treatment by selectively targeting specific bacterial pathogens. Vaccines are used to prevent disease by triggering a specific immune response only to those pathogens included in the vaccine. The unique composition of this immunomodulator is intended to rapidly trigger an animal's innate immune system, preparing it to better fight infection when challenged by BRD. Zelnate™ DNA Immunostimulant is a bacterial-produced plasmid DNA with a liposome carrier that stimulates the innate immune system in cattle. Per the label claim, Zelnate™ is indicated for use as an aid in the treatment of BRD due to *Mannheimia haemolytica* in cattle 4 months of age or older, when administered at the time of or within 24 h after a perceived stressful event. A summary of the technical studies conducted for regulatory approval by the USDA-CVB can be found at Reference 14.

As a bacterially produced plasmid DNA prepared with a liposome carrier, Zelnate™ is believed to act through binding of toll-like receptor (TLR)9. TLRs are evolutionarily conserved protein receptors that detect pathogen-associated molecular patterns and are key regulators of both innate and adaptive immune responses (for a brief review on TLRs, see 15). TLR9 binds unmethylated CpG DNA produced by bacterial, protozoal, and some viral pathogens. As such, bacterial CpG DNA and synthetic oligodeoxynucleotides containing unmethylated CpG are potent inducers of the innate immune system, including dendritic cells, macrophages, and natural killer cells. According to product literature, Zelnate™ significantly reduces lung lesion scores and mortality rates in cattle intratracheally challenged with *M. haemolytica*. DNA immunostimulants are a new way to fight BRD by jumpstarting the innate immune system and enhancing the body's own ability to respond quickly and effectively.

2.3. Imrestor™ Approved in 2014

Imrestor™ was approved initially in Mexico and subsequently in several central and South American countries. EU authorization was granted in 2015, followed in 2016 by New Zealand, then the United States and Canada, with simultaneous approval by the US Food and Drug

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Administration (FDA) and Health Canada. Administration of recombinant cytokines to modulate immunity in immunocompromised hosts has been used to prevent bacterial infections in humans (16). Recombinant human granulocyte colony stimulating factor (G-CSF) has been used successfully for many years as an adjunct therapy for neutropenic cancer patients undergoing chemotherapy. At the time of publication, only one cytokine for use as an immunomodulator has received simultaneous FDA and Health Canada approval for use in animal agriculture. This immunomodulator (Imrestor™) is a modified form of recombinant bovine granulocyte colony stimulating factor (rbG-CSF). The proprietary modification involves a targeted attachment of a polyethylene glycol moiety to a specific amino acid in the recombinant protein that extends the half-life of the rbG-CSF and is designed to have minimal to no interference with G-CSF binding to its receptor on bone marrow myeloid stem cells. Pegylation is thought to reduce renal clearance by the targeted protein having a higher molecular weight and larger hydrodynamic radius than the parent molecule. Thus, the protein is cleared from the body at a much slower rate and may reduce proteolytic degradation (17). This targeted pegylation technology therefore increases potency or specific activity and extends the duration of activity of an administered dose, making it more practical for use in its intended target species. The generic product name is pegbovigrastim, and it was approved in 2016 by the FDA and Health Canada as an immune restorative in postpartum dairy cows to aid in reducing the incidence of clinical mastitis in the first month of lactation. It is the first and only approved immune restorative for periparturient dairy cows and heifers. Per the approved label claim, pegbovigrastim reduces the incidence of clinical mastitis by 28% in the first 30 days of lactation in dairy cows and heifers. Peer-reviewed studies describe the mechanism of action of pegbovigrastim and report an even greater reduction (35%) in clinical mastitis incidence during the first 30 days of lactation in 4 studies conducted in the United States (18–21).

Although patent protection on rbG-CSF had expired years before the launch of pegbovigrastim, its commercial viability was enabled through licensing of the proprietary pegylation technology. This was a practical need for farm use of the product, because repeated daily injections in a dairy production setting would be impractical and cost prohibitive for dairy farmers. As a long-acting form of a cytokine that is naturally part of a cow's immune system, pegbovigrastim was determined to require no milk discard or slaughter withholding time following its administration. Pegbovigrastim works by increasing bone marrow production and output of neutrophils into the bloodstream at a time when a cow's immune system is functionally impaired due to physiological changes associated with giving birth. Hence, the innate immune response provided by an increased circulating pool of neutrophils is greatly enhanced.

To ameliorate the effects of periparturient immunosuppression, several studies evaluated various cytokines that are part of the cow's normal immune system (4, 22–27). Neutrophils are recognized as the primary immune defense against bacterial infections that cause mastitis. In a series of independent studies, G-CSF was evaluated for its effects on bovine immunity and as a prophylactic against mastitis pathogens (28–33). The original hypothesis for mastitis prevention in dairy cattle was that a sufficiently large pool of circulating neutrophils ready to launch an effective acute inflammatory response would reduce the chance of infection and clinical mastitis caused by experimental challenge in the bovine mammary gland. A transient delay in mounting a sufficient inflammatory response of neutrophils in the gland of cows shortly after calving compared with cows in mid-lactation had been reported to result in peracute disease and sometimes death (34, 35). In postpartum cows, that delay results in approximately 100× more *Escherichia coli* at peak infection and a much greater level of endotoxin produced compared with mid-lactation cows. Other studies have found mastitis severity to be inversely correlated with a rapid inflammatory response to intramammary infection (36–38). Early studies in cows with human or bovine G-CSF induced a dose-dependent neutrophilia between 30,000 and 60,000 neutrophils/μl, which

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in turn afforded a more rapid inflammatory response. Early bacterial challenge studies with G-CSF as a prophylaxis revealed that G-CSF was beneficial for reducing the incidence of *S. aureus* and *Klebsiella pneumoniae* mastitis (33, 39). However, as a therapeutic in cows with chronic *S. aureus* infections, no benefit was observed (40). Based on the initial studies indicating no significant adverse effects, a subsequent postpartum challenge pilot study demonstrated that G-CSF reduced the incidence and severity of clinical coliform mastitis by 50% during the first week of lactation following experimental challenge (see **Supplemental Appendix 2**; 41). Many studies using pegbovigrastim in periparturient dairy cows have been published subsequently, with most findings further supporting its label usage (20, 21, 42–66).

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Although the commercial use of rbG-CSF was approved and proven to work as an immune restorative on commercial dairy farms, it was challenging to support as a product due to cost-of-manufacturing issues, as well as for producers and technical support personnel to understand how the product works in comparison to vaccines or antibiotics. Being a product of biotechnology also brought resistance for its usage among some customers. These factors have contributed to the company no longer marketing Imrestor™ despite its efficacy in reducing mastitis incidence and antibiotic usage on farms that used it. Today, people ask whether Imrestor™ can be relaunched. The short answer is yes, it is possible; however, the likely corporate answer right now is no. First and foremost, among challenges that must be overcome is the cultural/societal stigma of it being a product of biotechnology. Second, knowing its dose-dependent efficacy in reducing disease and antibiotic usage in dairy cows, a next-generation product would benefit from an even greater increase in the neutrophilia and duration of effect. The approved dose of pegbovigrastim induced a sustained neutrophilia well below those observed in the early studies of the 1980s and 1990s that used multiple daily injections. With the ever-increasing global demand for animal protein for human consumption, the need for this immunomodulator to improve dairy cow health, welfare, and productivity may someday outweigh public hesitancy toward biotechnology products.

Finally, understanding what an immunomodulator does requires a significant understanding of immunology. It cannot be emphasized enough that pegbovigrastim is not an antibiotic or a vaccine. Some sectors of the dairy industry (and even some regulatory agencies) struggled to understand these distinctions and had misperceptions on how the product was expected to perform. Pegbovigrastim was a unique approach to reducing the incidence and/or severity of clinical mastitis. Vaccines take a few weeks before the host develops a sufficient level of protective immunity—often requiring a priming and subsequent booster dose 2–3 weeks apart. Very few vaccines ever effect sterilizing immunity, but they help prevent the incidence of and/or reduce the severity of disease caused by an infection with a specific pathogen. Antibiotic therapy works via direct bactericidal or bacteriostatic effects on susceptible pathogens. The extent to which antibiotics are effective is dictated by the dose administered and the pharmacokinetic and pharmacodynamic properties of each antibiotic. In the case of mastitis and the bovine mammary gland, the question becomes, Will the antibiotic, if parenterally administered, have a pKa to effectively ion trap in lacteal secretions that have a lower pH than serum/plasma? Alternatively, if administered through a teat cistern, will it effectively distribute up into the mammary gland and parenchymal tissues to attack the bacterial infection in all areas of the mammary gland? Both issues are large hurdles for antibiotic therapy to overcome.

As an immunomodulator, pegbovigrastim works through an innate immune mechanism in the bone marrow to elicit production of a greater number of neutrophils and their maturation and release into the circulating pool of neutrophils. The effect on the bone marrow takes a few days to reach the maximal neutrophilia. These neutrophils can perform innate immune surveillance in search of a tissue infection and then mount an effective inflammatory response wherein the antibacterial mechanisms of the neutrophil work to eliminate the bacteria and degrade microbial

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toxins that may have been produced. Thus, the antibacterial effect is mediated by increasing the host's available pool of neutrophils, which then have a greater capacity to kill the mastitis pathogens and degrade bacterial toxins that might be produced (67, 68). The efficacy therefore depends on the time to reach an effective neutrophilia and duration of the bone marrow response to pegbovigrastim in the host.

2.4. Amplimune®

This intravenously administered product was first licensed in 1993 as a mycobacterial cell wall fraction with immunostimulant activity. A study designed to demonstrate efficacy of immunotherapeutic treatment of 1- to 5-day-old calves facing an oral challenge with a K99 strain of *E. coli* was submitted to USDA-APHIS (Animal and Plant Health Inspection Service) for evaluation prior to licensure. At the time of approval, this product met the regulatory standards for acceptance. Based on a June 8, 2023, USDA summary of studies supporting product licensure, no data were published because the study was submitted to USDA-APHIS prior to January 1, 2007, and APHIS requires publication only of data submitted after that date.

Today, Amplimune® is owned by NovaVive, a private company founded in July 2014 that then acquired the *Mycobacterium* Cell Wall Fraction (MCWF) technology platform that was formerly part of Bioniche Animal Health. NovaVive (**http://novavive.ca/about-us**) reports to have a veterinary immunotherapeutic platform based on the MCWF technology with five regulator-approved products in the United States (three of which are regulator approved in Canada, two in Australia, three in New Zealand, and two in the United Arab Emirates). Amplimune® is an emulsion of MCWF that is regulator approved in the United States, Canada, New Zealand, and the United Arab Emirates to reduce clinical signs and mortality associated with *E. coli* K99 diarrhea in neonatal calves. When injected into the animal, Amplimune® is reported to enhance both innate and adaptive immune responses to fight bacterial infections without the use of antibiotics.

The Organic Materials Review Institute (OMRI) is an international nonprofit organization that determines which products are allowed for use in organic production and processing. OMRI Listed® products are allowed for use in certified organic operations under the USDA National Organic Program. Amplimune® is reported by NovaVive to be OMRI listed in the United States and Canada for use in organic production. NovaVive reports that Amplimune® has research demonstrating safety and efficacy for improving average daily gain in hutch-reared calves, reducing mortality and drug use in veal calves, reducing the risk of pneumonia in shipped calves, improving pregnancy in embryo transfer–recipient heifers, reducing disease/improving fertility of dairy cows, reducing morbidity and mortality in highly stressed feedlot calves, and enhancing efficacy as an adjunct therapy alongside traditional treatments.

3. WHAT OTHER IMMUNOMODULATORS ARE BEING RESEARCHED FOR ANIMAL HEALTH APPLICATIONS?

Future immunomodulator products in corporate pipelines are topics of speculation and often not what a company is actually conducting research on, because it is in the company's best interest to keep such information confidential until a product has been approved by the appropriate regulatory agency. However, a large body of peer-reviewed literature covering a range of immunomodulators is available. We highlight potential modulators of innate immunity below.

3.1. β-Glucans

β-Glucans (BGs) are polysaccharides of D-glucose monomers linked by β-glycosidic bonds and are a main constituent of the cell wall of some plants, fungi, bacteria, and yeast. In food animals,

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BGs typically are sourced from yeast and may be supplemented as fermentation products. BG are generally recognized as safe; therefore, they require no regulatory approval for use as a supplement in animal agriculture. Monocytes, dendritic cells, macrophages, and neutrophils express receptors including dectin-1 (a natural killer cell receptor-like C-type lectin thought to be involved in innate immune responses to fungal pathogens) and a β_2 integrin known as complement receptor 3. Dectin-1 and complement receptor 3 can recognize BG in its various forms, though their contribution to response can be cell type dependent (reviewed in 69). BG can be directly immunostimulatory and also induces innate memory (training/tolerance), which relies on epigenetic modification of immune cells for increased responsiveness to subsequent agonist exposure (reviewed in 70).

Bovine monocytes produce proinflammatory cytokines after exposure to particulate BG and upregulate expression of costimulatory molecules (71). BG can induce innate training in bovine monocytes, as monocytes exposed to *Saccharomyces cerevisiae* cell wall BG have heightened production of tumor necrosis factor after subsequent exposure to lipopolysaccharide (LPS), a major cell wall component of gram-negative bacteria such as *E. coli* (72). In newborn calves, oral BG administration leads to a trained phenotype in circulating monocytes, suggesting administration may improve protection from bacterial infection (73). Other in vivo trials have been performed, but they used yeast cell wall fractions or fermentation products, not purified BG. In these studies, it is unclear if BG is the causative component, but results are worth noting. Specifically, *S. cerevisiae* fermentation product (SCFP) supplementation enhances peripheral innate immune function in preweaned calves (74) and lessens the severity of infection with bovine respiratory syncytial virus (BRSV) as well as coinfection with BRSV and *Pasteurella multocida* (75). SCFP also reduces salmonellosis in preweaned dairy calves (76). The exact mechanism of SCFP effects outside the gastrointestinal tract is unknown, but dietary BG can cross the gastrointestinal barrier and likely bind respective BG receptors on circulating and/or tissue-resident cells to modify immune responses (74, 77).

BG recognition by porcine innate immune cells and subsequent immune response is cell-type (78, 79) and BG-type (80) specific. Although swine monocytes produce inflammatory cytokines in response to zymosan, a BG prepared from yeast cell wall, it results in reduced cytokine production after subsequent LPS exposure (innate tolerance) in vitro (81). BG type differentially impacts porcine monocyte-derived dendritic cell maturation and function, suggesting careful choice would be necessary for targeted immunomodulation (82). Treatment of porcine alveolar macrophages with BG induces IFN-α production and reduces African swine fever virus replication (83). In vivo, results are mixed on BG's ability to reduce disease through either prophylaxis or therapy. BG therapy reduces rotavirus disease severity (84) and reduces *Salmonella enterica* serovar Typhimurium abundance in the cecum (85). Dietary BG of different origins fed to pigs for two weeks after weaning reduced fecal shedding of enterotoxigenic *E. coli* and diarrhea upon experimental challenge (86). Vaccination studies were unaffected by dietary BG supplementation (87–89); however, soluble BG augments T cell IFN-γ production in an antigen-independent manner (90).

Overall, the use of BG in food animals [including fish (91, 92) and poultry (93)] presents a promising approach for enhancing animal health via immunomodulation. However, several considerations must be addressed for effective use. As noted, the source and molecular structure of BG can significantly affect its immunomodulatory properties. Additionally, BG molecular weight and solubility can influence their bioactivity and efficacy as immunomodulators. The optimal dosage and administration method of BG are also critical factors that must be determined for each animal species. Therefore, more research is needed to establish standardized guidelines for use of BG in food animal production.

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3.2. Pattern Recognition Receptor Agonists

Use of pattern recognition receptor agonists to engage innate immunity has been explored most widely in poultry. In early studies, in ovo treatment with the TLR7 agonist loxoribine inhibited influenza A infection in 10-day-old embryonated eggs (94). In more recent studies, combinations of unmethylated CpG oligodeoxynucleotides (CpG) and polyinosinic:polycytidylic acid given in ovo protected chicks from yolk sac infection with avian pathogenic *E. coli* both pre- and post hatch (95). Delivery of CpG ODN in ovo also reduced infectious bronchitis virus loads in eggs prehatch and reduced chick morbidity and mortality post hatch (96). Use of TLR agonists for immune modulation has been less explored in swine and cattle. Zelnate™ was formulated as a liposome encapsulating CpG-rich DNA plasmids. In recent work, the same group that developed this technology explored a second-generation immunostimulant, liposome-TLR complexes or LTC, composed of liposomes encapsulating dual TLR3 and TLR9 agonists (97). Intranasal treatment of weaned Holstein calves with LTC reduced clinical disease and disease-associated mortality in a seeder challenge model (97). In vitro stimulation of peripheral blood mononuclear cells from cattle, swine, and poultry has shown that LTC induces robust type I IFN responses and nitric oxide production in all three species (98). Although this avenue of research is promising, the safety and toxicity of pattern recognition receptor agonist–based formulations must be considered carefully. Similar to cytokine administration, there is a narrow window of safety and potential for unwanted side effects.

3.3. Recombinant Host Defense Peptides

Host defense peptides (HDPs) are a broad class of bioactive molecules with both antimicrobial and immunomodulatory activities. Although their antibacterial activity against an array of animal pathogens is a significant benefit (99–105), their effects on host immunity may be more extensive and relevant in vivo (106, 107). HDPs can act as chemoattractants and mediate pro- and anti-inflammatory effects (107). Oral treatment with microcin, an antimicrobial peptide produced by *E. coli*, modulates proinflammatory cytokine production, increases serum IL-10 concentrations, and reduces episodes of diarrhea in weaned pigs (108). In chickens, in ovo administration of recombinant avian β-defensin-1 reduced the mortality caused by experimental avian pathogenic *E. coli* yolk sac infection (109). HDPs have been studied extensively as alternatives or additions to traditional antibiotics, and a few have advanced to clinical trials; however, none has been granted regulatory approval to date (106). Practical use of HDPs in animal or human health has been limited due to the high cost of production and scale-up, requiring either chemical synthesis or purification from natural sources. However, HDPs can be produced more cost effectively with recombinant DNA technology, using yeast, bacteria, or even mammalian cells as expression systems. Recent advancements in the use of combinations or multidomain HDPs containing fusions of several synergistic HDPs have also been shown to increase the efficacy of HDPs while driving down production costs (106, 107). Thus, this technology may be promising for future animal health applications.

3.4. Immunobiotics

The general term probiotic was developed to define microbes that have beneficial effects on health. The term immunobiotics has been adopted to refer to microbes that have beneficial effects on the innate and adaptive immune system (110). Several immunobiotic strains have been shown to positively impact gut health and reduce the negative impact of enteric infection both in vitro and in vivo (reviewed in 111). For example, supplementation with *Lactobacillus rhamnosus* reduced clinical disease and improved gut barrier function in weaned piglets infected with porcine rotavirus (112).

Feeding a combination of *Lactobacillus acidophilus* and *Lactobacillus reuteri* regulated recruitment of inflammatory myeloid cells to the gut, limiting immunopathology during rotavirus challenge in preweaned piglets (113). *Ligilactobacillus* and *Bifidobacterium* strains have also been shown to modulate inflammatory responses in porcine intestinal epithelial cells exposed to the TLR3 agonist polyinosinic:polycytidylic acid (114). In cattle, both oral supplementation and intramammary infusion of immunobiotics have been explored for reducing or preventing mastitis (reviewed in 115). In one in vivo study, intramammary infusion of *Bifidobacterium breve* enhanced innate immune responses in the mammary tissue of cows with chronic, subclinical mastitis (116). In vitro treatment of bovine mammary epithelial cells with various strains of *Lactobacillus* modulated LPS-induced inflammatory responses (117, 118), suggesting immunobiotics may both enhance and modulate innate immunity, depending upon the model explored. Interest in immunobiotic treatments is likely to increase because these approaches are cost effective and may be given in feed, although challenges with dosing and shelf life may limit widespread use.

4. FUTURE OF IMMUNOMODULATORS FOR ANIMAL AGRICULTURE

Given the cost of bringing an immunomodulator to the marketplace, there must be a significantly large market opportunity for a company to invest in developing such a product. The most expensive diseases affecting livestock include respiratory disease (e.g., pneumonia), enteric disease (diarrhea), and inflammation of the mammary gland (mastitis). The largest markets are with cattle and swine. The profit margins for poultry production are incredibly narrow and require great ease of use (mass administration).

One philosophy of immune modulation is to modulate a specific weakness of the immune system. With pegbovigrastim, the concept was to increase the available pool of circulating neutrophils that are responsible for protecting the mammary gland of the cow during a period of well-documented immune suppression (e.g., delayed inflammation, decreased phagocytic killing capacity, impaired IFN-γ production, and a failure to produce typical anamnestic responses to vaccine boosters). Similarly, weaning, comingling from various source farms, and shipping transport stress of beef cattle will cause a similar broad-based immune suppression. Other production phases of the livestock life cycle have impaired IFN-γ production (119, 120). One example of this is newborn animals, in which various aspects of the immune system are impaired or underdeveloped (121, 122) and some functional aspects may take until puberty to reach the full functional capability seen in adult animals (123). Impaired IL-18 production has been reported in neonatal piglets as a potential contributor to opportunistic disease (124). Each of these scenarios is worthy of pursuit because antibiotic use is under severe regulatory pressure, and very few new antibiotics for use in veterinary medicine are likely to ever be approved by the FDA. Many pharmaceutical companies have abandoned the pursuit of new antibiotics for use in veterinary medicine because of regulatory uncertainty of approval. Hence, immunomodulators that effectively target specific immune deficiencies could provide a new avenue for supporting the production of healthy livestock. Research is needed to identify the key underlying impairment of immunity and how to best restore immune function to an effective level. Such immune deficiencies will require a delicate balance in restoring the immune system without contributing to immune pathology or acute immune toxicity with some cytokines that have a very narrow dose margin of safety.

5. NUTRITIONAL APPROACHES TO IMMUNE MODULATION

Dietary immune treatments are also an area of intense investigation. Although not a major focus of this review, considerable research has been conducted on the topic, and managing optimal nutrient levels with ingredients such as vitamin E and selenium is well recognized to minimize immune

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impairment associated with nutrient deficiencies (125–132). However, little evidence supports hyper-supplementation of nutrients to enhance immune function. Metabolic disorders contribute to increased disease incidence in transition dairy cows. One study demonstrated that prevention of parturient hypocalcemia or milk fever by parathyroid hormone administration failed to alter the development or duration of immune suppression in transition cows (133). A subsequent experiment with mastectomized cows revealed that the effects of lactation contribute to the duration of immune suppression after calving but that the onset of immune suppression still occurs (134). That study concluded that hypocalcemia in the transition cow is a result solely of the calcium drain of lactation. Moreover, the act of parturition affected blood phosphorus, dry matter intake, and nonesterified fatty acid concentrations independent of the effect of lactation. A recent study concluded that cows with subclinical hypocalcemia (SCH) have fewer circulating neutrophils during the first 3 days of lactation, and a reduced proportion of these neutrophils demonstrate phagocytic activity. However, oxidative burst activity by neutrophils did not differ between cows with SCH and normocalcemic controls (135). This same study demonstrated that cows with SCH are at a higher risk of uterine disease and have higher serum levels of nonesterified fatty acids and β-hydroxybutyrate in the first 12 days of lactation. The authors suggested that SCH increases lipid stores mobilization and compromises neutrophil function, thus increasing the risk of uterine diseases, resulting in a reduced rate of pregnancy and extended days to conception later in lactation. Clearly, prevention of parturient hypocalcemia is critical to optimal performance of dairy cows, even if the effect of hypocalcemia on immune function varies by study.

Negative energy balance in postpartum dairy cows results when the demands of lactation are not met from diet consumed, necessitating mobilization of body fat stores, and contributes to the duration of postpartum immune suppression (134, 136, 137). However, experiments designed to maintain dietary energy intake have had mixed results on enhancing the immune capacity of transition cows (138). Negative protein balance in transition cows is another recognized nutritional challenge. Transition cows likely require greater amounts of methionine for liver and immune function but also for milk protein synthesis after calving. In a study examining the effects of methionine availability during the transition period on postpartum performance and neutrophil function (139), the authors concluded that methionine plays an important role in immune function, but the neutrophil test they used in their study did not measure bacterial killing capacity of neutrophils. Instead, the test they used, as described by the kit manufacturer, measures the overall percentage of neutrophils undergoing phagocytic uptake of an already-killed bacteria and the number of bacteria taken up per neutrophil. In our previous immune suppression studies (133, 140, 141), we routinely observed that phagocytic uptake of bacteria by bovine neutrophils often increases, while paradoxically, oxidative killing is impaired. Under these circumstances, neutrophils may ingest greater numbers of bacteria because they have not spent their energy stores on oxidative killing. Hence, increased ingestion in the absence of functional killing may not be immune enhancing.

Immunomodulatory feed ingredients have also received considerable research interest regarding possible beneficial effects on immunity and health in dairy cows. One such product, OmniGen AF, has several published studies reporting to enhance innate immunity parameters and increase milk production in dairy cows (142–147). However, as of this writing, neither the USDA-CVB nor the FDA has approved it as an immunomodulator.

6. EFFECTS OF IMMUNOMODULATORS ON NORMAL PHYSIOLOGY

Immunomodulators would be expected to interact with normal metabolism; however, disease itself has a metabolic expense resulting from diverting energy, protein, and minerals to sustaining the immune response that may far exceed what an immunomodulator may elicit. There is also an

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inherent increase in the basal metabolic rate associated with a febrile response during disease. Ideal immunomodulators will not cause injection site reactions and fever. Studies on the metabolic changes associated with induction of a systemic inflammatory response in humans with severe sepsis are reported to increase resting energy expenditure progressively over 7 days to ~40% above normal and may remain elevated 3 weeks from the onset of illness (148). In addition, over a 3-week period, those patients lost 13% of their total body protein. The metabolic cost/benefit ratio of preventing a disease running through a herd or flock versus the metabolic cost of receiving an immunomodulator is a factor that would affect market penetration of an immunomodulator. This type of research is woefully lacking in livestock and would help demonstrate the return on investment for producers. However, these studies require metabolism cages and are very difficult to conduct on large scales.

Regarding other possible endocrine effects, a critical component of regulatory approval for pegbovigrastim was to demonstrate no negative impact on pregnancy or lactation. These regulatory studies might be required for commercial approval, and the outcome may vary by immunomodulator type.

7. FACTORS IMPACTING ANY FUTURE IMMUNOMODULATORS FOR ANIMAL HEALTH

Regardless of which immunomodulator is developed or how it mediates its effects, there are pragmatic aspects for it to be a successful tool for animal agriculture. As such, it needs to work within the normal practices of each livestock or poultry production system and preferably not require excessive animal handling. In livestock and poultry production, every sector involved with the use of an immunomodulator must make a return on the financial investment. This is true for the end user (producer) as well as the businesses that manufacture, market, and distribute or prescribe the product.

7.1. Return on Investment for a Company

A significant factor impacting the commercial viability of immunomodulators as a class of compounds is the cost of approval by a regulatory agency and the ongoing pharmacovigilance that is required. A reasonable research and development cost estimate for an FDA-approved immunomodulator ranges from $10 million to $30 million for studies supporting regulatory approval, supplies, and a prorated portion of the salaries of company personnel based on their level of involvement. FDA sponsor fees (~$600K) or other regulatory charges are additional expenses that vary by regulatory agency. If the product or enabling technologies are licensed from another firm, those costs would increase but would be highly variable based on how advanced the product development is at the time of licensing and perceived market value. USDA registrations would be at the lower end of this range and would include monoclonal antibodies because they are eligible for USDA registration. With USDA-APHIS-CVB, a conditional approval is possible that would enable selling the product while completing the work for the full approval. The FDA can also designate a product as a minor use/minor species product like an orphan drug. This provides a sponsor with the ability to obtain conditional approval and start sales once they submit the human/target animal safety and Good Manufacturing Practice package with evidence of likely efficacy. A significant reduction in fees is possible with this latter system.

7.2. Return on Investment for Livestock and Poultry Producers Is Essential

An immunomodulator for use in veterinary medicine cannot cost what a similar product for use in humans would cost. Therefore, companies must minimize drug discovery and development

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research costs, and manufacturing processes must be extremely efficient and avoid excessive transportation transfers of materials during the manufacturing and packaging processes. In addition, in large herd or flock systems, multidose packaging is essential for practical use reasons on the farm. Not addressing these points will significantly increase the cost of goods, making the selling price for a disease preventative higher than otherwise necessary and possibly making it cost prohibitive for most producers. Most importantly, livestock producers must be able to see improvement in the health of their herd or flock without the need for intricate record keeping and analysis. The latter criterium is perhaps the tallest hurdle to clear for an immunomodulator. Equally critical to this last point is that producers must understand the incidence and cost of a disease that a product is designed to prevent or reduce. A recent randomized blind study of the economic benefit of pegbovigrastim involving 2,153 cows in 4 separate commercial grazing dairy herds in Uruguay (controls: 1,114 cows; pegbovigrastim: 1,039 cows) reported a statistically significant benefit of pegbovigrastim administration during the transition period of $210 partial net return per cow per year (64). This economic return was based on milk revenues and costs for feed, medical treatments, insemination, and culling over a full lactation. Costs of pegbovigrastim (2 doses) and its application were not included in the economic calculations, hence resulting in an approximate 6:1 return on investment to the producer based on $35 for 2 doses of pegbovigrastim. Labor costs associated with the two doses were not estimated due to between-farm variation in labor availability and cow-handling facilities. However, labor costs would not be expected to exceed the cost of pegbovigrastim. Finally, marketing expectations of an immunomodulator must also remain within expectations of what the product will do for the animal's health. It is always possible that a product may outperform expectations, but the label claim is what the product must be promoted on.

SUMMARY POINTS

1. Immunomodulators must enhance immunity without adverse effects.
2. Immunomodulators must be easy to use and fit into modern production practices.
3. Regulatory approval is required, and agency oversight varies within and between countries.
4. Both companies and producers must have a sufficient return on their investment to justify the marketing and use of immunomodulators on farms.

FUTURE ISSUES

1. Costs associated with regulatory approvals are substantial and impact the likelihood of new product research and development.
2. Oral delivery in feed or water may be the most economical means of using immunomodulators in livestock.

DISCLOSURE STATEMENT

M.E.K. was an external reviewer of the regulatory safety and efficacy studies submitted for the European Medicines Agency recommendation on Imrestor™ approval in the European Union. The authors are not aware of any affiliations, memberships, funding or financial holdings that might be perceived as affecting the objectivity of this review.

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